

GEE estimator $\hat{\theta}$ is the solution of

$$S(\theta) = \sum_{k=1}^K S_k(\theta) = \sum_{k=1}^K D_k^T(\theta) V_k^{-1} (Y_k - \mu_k(\theta)) = 0$$

$$0 = S(\hat{\theta}) \approx S(\theta) + \frac{\partial S(\theta)}{\partial \theta} (\hat{\theta} - \theta)$$

$$\hat{\theta} - \theta \approx - \left[\frac{\partial S(\theta)}{\partial \theta} \right]^{-1} S(\theta)$$

$$\sqrt{K}(\hat{\theta} - \theta) \approx - \left[\frac{1}{K} \sum_{k=1}^K \frac{\partial S_k(\theta)}{\partial \theta} \right]^{-1} \left[\frac{1}{\sqrt{K}} S(\theta) \right] \text{ ----- } \textcircled{1}$$

$$- \frac{1}{K} \sum_{k=1}^K \frac{\partial S_k(\theta)}{\partial \theta} = - \frac{1}{K} \sum_{k=1}^K \frac{\partial D_k^T(\theta)}{\partial \theta} V_k^{-1} (Y_k - \mu_k(\theta)) - \frac{1}{K} \sum_{k=1}^K D_k^T(\theta) V_k^{-1} \left(- \frac{\partial \mu_k(\theta)}{\partial \theta} \right)$$

$$\approx \frac{1}{K} \sum_{k=1}^K D_k^T(\theta) V_k^{-1} D_k$$

$$\begin{aligned} \text{Since } \frac{1}{K} \sum_{k=1}^K \frac{\partial D_k^T(\theta)}{\partial \theta} V_k^{-1} (Y_k - \mu_k(\theta)) &\xrightarrow{P} E \left(\frac{\partial D_k^T(\theta)}{\partial \theta} V_k^{-1} (Y_k - \mu_k(\theta)) \right) \\ &= \frac{\partial D_k^T(\theta)}{\partial \theta} V_k^{-1} E(Y_k - \mu_k(\theta)) \\ &= 0 \end{aligned}$$

Then

$$\sqrt{K}(\hat{\theta} - \theta) \approx \left[\frac{1}{K} \sum_{k=1}^K D_k^T(\theta) V_k^{-1} D_k(\theta) \right]^{-1} \left[\frac{1}{\sqrt{K}} S(\theta) \right]$$

$$\text{So } \text{cov}(\sqrt{K}(\hat{\theta} - \theta)) \approx \left[\frac{1}{K} \sum_{k=1}^K D_k^T(\theta) V_k^{-1} D_k(\theta) \right]^{-1} \text{cov} \left(\frac{1}{\sqrt{K}} S(\theta) \right) \left[\frac{1}{K} \sum_{k=1}^K D_k^T(\theta) V_k^{-1} D_k(\theta) \right]^{-1}$$

$$\begin{aligned}
\text{Cov}\left(\frac{1}{\sqrt{K}}S(\theta)\right) &= \text{Cov}\left(\frac{1}{\sqrt{K}} \sum_{k=1}^K D_k^T(\theta) V_k^{-1} (Y_k - p_k(\theta))\right) \\
&= \frac{1}{K} \sum_{k=1}^K \text{Cov}\left(D_k^T(\theta) V_k^{-1} (Y_k - p_k(\theta))\right) \\
&= \frac{1}{K} \sum_{k=1}^K D_k^T(\theta) V_k^{-1} \text{Cov}(Y_k - p_k(\theta)) V_k^{-1} D_k(\theta)
\end{aligned}$$

Note that $\text{Cov}(Y_k - p_k(\theta)) = \text{Cov}(Y_k) = V_k$.

$$\text{So } \text{Cov}\left(\frac{1}{\sqrt{K}}S(\theta)\right) = \frac{1}{K} \sum_{k=1}^K D_k^T(\theta) V_k^{-1} D_k(\theta)$$

Then we have

$$\begin{aligned}
\text{Cov}(\sqrt{K}(\hat{\theta} - \theta)) &\approx \left[\frac{1}{K} \sum_{k=1}^K D_k^T V_k^{-1} D_k \right]^{-1} \left[\frac{1}{K} \sum_{k=1}^K D_k V_k^{-1} D_k \right] \left[\frac{1}{K} \sum_{k=1}^K D_k V_k^{-1} D_k \right]^{-1} \\
&= \left[\frac{1}{K} \sum_{k=1}^K D_k^T V_k^{-1} D_k \right]^{-1}
\end{aligned}$$

$$\text{So } \text{Cov}(\hat{\theta}) \approx \left[\sum_{k=1}^K D_k^T V_k^{-1} D_k \right]^{-1}$$